

1. Administrative & Study Identification

Full Study Title: An Open-label Dose Escalation and Expansion, Followed by a Phase II Study of Tulumetostat (DZR123) and JSB462 (Luxdegalutamide) in Patients With Progressive Metastatic Castrate Resistant Prostate Cancer (mCRPC) **Short Title/Acronym:** TumiSTAR-01 **Protocol Number:** CDZR123A12107 **NCT Number:** NCT07206056 **Sponsor Name:** Novartis **Investigational Product:** Tulumetostat (DZR123) and JSB462 (Luxdegalutamide) **Phase of Development:** Phase I/II **Medical Monitor:** Novartis Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: * Phase 1a: To determine the recommended dose(s) for expansion (RDE), safety, and tolerability. * Phase 1b: To determine the safety, tolerability, recommended Phase 2 dose (RP2D), and PSA50 response at 6 months at the RDEs. * Phase 2: To evaluate PSA50 at 6 months comparing the combination therapy vs. Standard of Care (SOC).

Secondary Objectives: To evaluate Pharmacokinetics (PK), radiographic Progression-Free Survival (rPFS), Overall Survival (OS), Objective Response Rate (ORR), Best Overall Response (BOR), Duration of Response (DOR), and Time to Symptomatic Skeletal Event (TTSSE).

2.2 Background & Rationale Prostate cancer is a leading cause of cancer death, and mCRPC remains difficult to treat due to acquired resistance to androgen receptor pathway inhibitors (ARPIs). Preclinical studies implicate EZH2 in mediating resistance to ARPIs. Combining AR-directed treatment (luxdegalutamide, a PROTAC AR degrader) with EZH2/EZH1 inhibition (tulumetostat) could potentially overcome this resistance and improve clinical outcomes for patients with progressive mCRPC.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Phase II compares to SOC: Choice of ARPI, chemotherapy, or lutetium vipivotide tetraxetan) **Blinding:** Open-label **Randomization:** 1:1 (Applicable in Phase 1b and Phase II)

3.2 Planned Enrollment & Duration Target **\$N\$:** 188 male participants **Number of Sites:** Global, multicenter **Estimated Study Start:** 15-Oct-2025 **Estimated Primary Completion:** 17-Dec-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult men ≥ 18 years of age. 2. Histologically and/or cytologically confirmed adenocarcinoma of the prostate without neuroendocrine or small cell features. 3. Evidence of mCRPC with ≥ 1 metastatic lesion present on screening/baseline imaging (CT, MRI, or bone scan) obtained ≤ 28 days prior to the start of treatment. 4. ECOG performance status 0-2 and life expectancy ≥ 6 months.

4.2 Exclusion Criteria 1. CNS metastases that are neurologically unstable or requiring corticosteroids for the purpose of maintaining neurologic integrity. 2. Prior taxane chemotherapy in the mCRPC setting (specifically excluded for Phase II participants; taxane-naïve required for Phase II, though prior chemo is allowed in Phase I). 3. Any unstable disorder or medical condition that can impact participant safety or study outcomes.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * Phase 1a: Tulumetostat QD + luxdegalutamide (100 or 300 mg). * Phase 1b: Tulumetostat (RDE) QD + luxdegalutamide (100 or 300 mg QD). * Phase 2: Tulumetostat (RP2D) QD + luxdegalutamide (100 or 300 mg QD) vs SOC. **Route of Administration:** Oral **Duration of Treatment:** Cycles continue until disease progression, unacceptable toxicity, or study completion.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of Care (Choice of ARPI, chemotherapy, or lutetium (^{177}Lu) vipivotide tetraxetan) as control arm treatments. Medications allowed for co-morbidities. **Prohibited:** Concomitant investigational agents, corticosteroids used to maintain neurologic integrity in CNS metastases, and therapies directly interfering with the study drugs.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Baseline Imaging (CT/MRI/Bone scan) Baseline
Day 0	Randomization (Phase 1b/II), First Dose, Safety Labs		
Interim	Periodic	Safety Assessments, PSA monitoring (Months 3, 6, 9, 12), PK sampling	End of Study End of Tx Final Vital Signs, Follow-up for OS, rPFS, and TTSSE

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: In Phase I: Recommended Dose for Expansion (RDE), Recommended Phase 2 Dose (RP2D), Safety and Tolerability. In Phase II: PSA50 (50% reduction in Prostate-Specific Antigen) at 6 months. **Measurement Method:** Clinical assessments, dose-limiting toxicity (DLT) evaluations, and central/local laboratory analysis for PSA levels.

7.2 Secondary & Exploratory Endpoints * Radiographic Progression-Free Survival (rPFS) and Overall Survival (OS) * Objective Response Rate (ORR), Best Overall Response (BOR), and Duration of Response (DOR) * Time to Symptomatic Skeletal Event (TTSSE) and Pharmacokinetics (PK)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via patient interviews and clinical evaluations. Dose escalation safety is guided by a Bayesian logistic regression model with overdose control. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External safety review committee overseeing the Bayesian logistic regression dose-escalation phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy endpoints; Safety Set for AE/SAE evaluations. **Sample Size Justification:** $N=188$, structured to establish MTD/RDE in Phase I and appropriately power the Phase II randomization comparing combination therapy vs. SOC. **Handling of Missing Data:** The time-to-event efficacy endpoints (rPFS, OS, TTSSE) will be analyzed using Kaplan-Meier estimates.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits per sponsor schedule. **Audit Trail:** eCRF locking and standardized data clarification forms.

11. Study Identifiers & Governance

Primary ID: CDZR123A12107 **Registry Number:** NCT07206056 **Sponsor/Lead Organization:** Novartis **Study Title:** TulmiSTAR-01 **Official Regulatory Title:** An Open-label Dose Escalation and Expansion,

Followed by a Phase II Study of Tulumimmetostat (DZR123) and JSB462 (Luxdegalutamide) in Patients With Progressive Metastatic Castrate Resistant Prostate Cancer (mCRPC) (TulmiSTAR-01)

12. Clinical Framework (Oncology Specific)

Primary Condition: Progressive Metastatic Castrate Resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** Histologically/cytologically confirmed prostate adenocarcinoma; metastatic lesion(s) present on imaging. **Medical Methodology:** Dual inhibition of EZH2/EZH1 combined with an AR degrader (PROTAC) vs. Standard of Care. **Optimization Target:** PSA50 reduction and radiographic progression-free survival.

13. Study Procedures & Methodology

Management Pathway: Targeted oncology pathway for mCRPC **Education Protocol (HCP):** Investigator protocol training, AE management, and Bayesian dose-escalation parameters. **Education Protocol (Patient):** Informed consent, dosing schedule education, and side-effect reporting. **Quality Audit Mechanism:** Clinical monitoring and centralized imaging/lab verification.

14. Observation & Follow-Up Schedule

Total Duration: Until completion (Estimated Dec 2030) **Onsite Visit Cadence:** Aligned with dose administration cycles and Phase I/II assessment timepoints (e.g., 3, 6, 9, 12 months for PSA) **Remote Monitoring Frequency:** Periodic safety follow-ups as required by the protocol. **Checklist Review Interval:** Routine check-ins for dose-limiting toxicities and survival follow-up.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				